RED OASIS | FIELD RESEARCH BRIEF

Shureefa • Oman pilot site, date-palm baseline, Mars greenhouse reference, and field-validation plan

PILOT SITE	Falaj al-Dariz palm oasis, Ad Dariz, Wilayat Ibri, Al Dhahirah, Oman	**COORDINATES**	23.308639 N, 56.613083 E
CROP	Date palm (*Phoenix dactylifera*), the dominant crop irrigated by aflaj in the study.	**AREA**	About 21.6 ha for the selected visible grove; rough map measurement, boundary needs GIS/field confirmation.
SATELLITE	Open the pinned site in Google Maps satellite view	**RATIONALE**	Arid oasis with published, site-specific evidence of date-palm irrigation demand exceeding supply.

1 | Earth crop and water baseline

Al-Ghafri's Falaj al-Dariz case study used observations from 2002–2003. Reported annual date-palm crop water demand (ETc) was **3,620 mm**; field supply including effective rainfall was **3,235 mm**. The demand/supply ratio was **1.12**: demand exceeded supply by 385 mm (about 10.6% of demand). Converted values are approximately **36,200 m3/ha/year** demanded versus **32,350 m3/ha/year** supplied. These are historical site-study values, not current measurements or a universal date-palm prescription. [1]

Fruit development follows flowering/pollination and five stages: **Hababouk** → **Kimri** → **Khalal** → **Rutab** → **Tamar**. Interpret irrigation alongside season, cultivar, palm age, soil, and irrigation method; confirm the current schedule with a grower before using it as an operational recommendation. [2]

2 | Mars mode: NASA evidence and prototype settings

NASA's MarsOASIS is a greenhouse concept for growing Outredgeous lettuce, with autonomous monitoring and a closed water-recycling system; it is not evidence that date palms have been grown on Mars. [3] NASA greenhouse research gives broad plant-growth bounds of about **10–35 C** and **40–85% relative humidity**. [4] For a reproducible demo profile, use a NASA ISS lettuce reference: **23.25 C, 16 h light / 8 h dark, 200 micromol m^-2 s^-1**. These are lettuce chamber settings, not validated Mars or date-palm setpoints. [5] Set water recovery ON; test external make-up water at **50%, 75%, and 100%** of the Earth baseline as team stress-test assumptions, not NASA-published Mars values.

3 | Published Oman comparison data

A separate 2015–2016 field experiment at Al Kamil and Al Wafi Agricultural Research Station tested Khalas palms. Reported annual water use was **9,966 m3/ha** with bubbler irrigation versus **6,421 m3/ha** with subsurface drip at 60% ETc (about 35.6% less); yield was **79.0 vs 78.3 kg/palm**, with no significant difference reported. At 40% and 20% ETc, yield fell to 69.8 and 68.8 kg/palm. These are published experimental results from South Al Sharqiyah, not measurements from Dariz or a farmer interview; use them as a comparison, not a direct site estimate. [6]

4 | Data-use note

These published datasets can replace an interview as the source for a documented comparison if the challenge accepts secondary research. Label them as published data; they do not count as a personal interview. Dariz values are from 2002–2003, while the irrigation trial is from South Al Sharqiyah in 2015–2016. A SMAP soil-moisture pixel is about 9 km across, far larger than this grove, so use it as regional context rather than field-level ground truth. [7]

REFERENCES

[1] Al-Ghafri, A. (2006). "Aflaj's Irrigation Water Demand/Supply Ratio: Two Case Studies." Journal of Agricultural and Marine Sciences, 11(1), 85–92.

[2] FAO. Date Palm: Botanical and Systematic Description — fruit growth stages. [3] NASA NTRS. Mars OASIS System Reference Manual.

[4] Wheeler (2000), NASA NTRS. Can CO2 be Used as a Pressurizing Gas for Mars Greenhouses?. [5] NASA OSDR, OSD-267, Veggie lettuce protocol.

[6] Dhehibi et al. (2022). Economic and environmental evaluation of different irrigation systems for date palm production in the GCC: Oman and Saudi Arabia. [7] NSIDC, SMAP L3 soil-moisture product.